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Department: B.E CSE

Subject: Web Interface Unit 05

Batch : 04

Project 02 : Student Progress Card using React

Aim

To develop a Student Progress Card web application using React that displays subject-wise marks in tabular form, and dynamically calculates the total marks, percentage, and overall grade using JavaScript logic embedded in a functional component.

Procedure

Step 1: Creating the React Application

A React application is created using Vite. The progress card is built as a single reusable component, Progress.jsx, styled with a dedicated Progress.css file.

Step 2: Defining the Subject Data

An array of subject objects, each holding a subject name and marks, is defined inside the component. This array drives the entire marks table, so the table is generated from data rather than being hard-coded row by row.

Step 3: Calculating Total and Percentage

The Array.prototype.reduce() method is used to sum all subject marks into a total. The percentage is derived by dividing the total by the number of subjects.

Step 4: Determining the Overall Grade

An if-else if-else chain checks the percentage against grade boundaries (90, 80, 70, 60) and assigns an overall grade of A+, A, B, C, or D accordingly.

Step 5: Rendering the Marks Table

The subjects array is rendered using Array.prototype.map(), producing one table row per subject. Each row calculates its own per-subject grade inline using nested ternary conditional rendering, and displays a PASS status.

Step 6: Displaying the Summary Section

A summary section shows the total marks, percentage, overall grade, and final result (PASS) in a four-column grid layout, computed from the values derived in earlier steps.

Step 7: Adding College Header, Student Details, and Footer

Static sections for the college name and department, the student's name and register number, and a footer with signature lines for the Class Teacher and Principal complete the printable progress card layout.

Step 8: Applying Styling and Responsive Design

CSS is used to style the header, table, and summary grid. A media query at 700px switches the student details to a stacked layout, reduces table padding and font size, and adjusts the summary grid to two columns for smaller screens.

Step 9: Testing the Application

The application is executed using the Vite development server. The app is tested by checking:

- Correct rendering of all subject rows from the data array
- Accuracy of the total, percentage, and grade calculations
- Per-subject grade logic against the marks table
- Responsive layout on smaller screen widths

React Concepts Used

- Functional Components
- JSX
- Array Methods (reduce, map)
- Conditional Rendering (if-else, ternary operator)
- Component-Based Architecture
- CSS Styling
- Responsive Web Design (media queries)

Code

Progress.jsx

```
import React from "react";
```

```
import "./Progress.css";
```

```
function Progress() {
```

```
  const subjects = [
```

```
    { name: "Web Interface", marks: 85 },
```

```
    { name: "Data Structures", marks: 78 },
```

```
    { name: "Database Management", marks: 82 },
```

```
    { name: "Java Programming", marks: 88 },
```

```
    { name: "Computer Networks", marks: 75 },
```

```
  ];
```

```
  const total = subjects.reduce((sum, subject) => sum + subject.marks, 0);
```

```
  const percentage = total / subjects.length;
```

```
  let grade = "";
```

```
if (percentage >= 90) {  
    grade = "A+";  
} else if (percentage >= 80) {  
    grade = "A";  
} else if (percentage >= 70) {  
    grade = "B";  
} else if (percentage >= 60) {  
    grade = "C";  
} else {  
    grade = "D";  
}
```

```
return (
```

```
<div className="progress-page">
```

```
<div className="progress-card">
```

```
    { /* College Header */ }
```

```
<div className="college-header">
```

```
<h1>PRINCE DR. K. VASUDEVAN COLLEGE OF ENGINEERING AND  
TECHNOLOGY</h1>
```

```
<p>Department of Cyber Security</p>
```

```
<h2>STUDENT PROGRESS CARD</h2>
```

```
</div>
```

```
    { /* Student Details */ }
```

```
<div className="student-details">
```

```
<div>
```

```
<p><b>Name:</b>Flana </p>
```

```
<p><b>Register No:</b> 411625104021</p>
```

```
</div>
```

```
<div>
```

```
<p><b>Department:</b> B.E.</p>
```

```

    <p><b>Semester:</b> III</p>
  </div>
</div>
{ /* Marks Table */}
<table>
  <thead>
    <tr>
      <th>S.No</th>
      <th>Subject</th>
      <th>Marks</th>
      <th>Grade</th>
      <th>Result</th>
    </tr>
  </thead>
  <tbody>
    {subjects.map((subject, index) => (
      <tr key={index}>
        <td>{index + 1}</td>
        <td>{subject.name}</td>
        <td>{subject.marks}</td>
        <td>
          {subject.marks >= 90
            ? "A+"
            : subject.marks >= 80
            ? "A"
            : subject.marks >= 70
            ? "B"
            : subject.marks >= 60
            ? "C"
            : "D"}
        </td>
        <td className="pass">PASS</td>
      </tr>
    )
  )}

```

```

    ))}
  </tbody>
</table>

  { /* Summary */
  <div className="summary">
    <div>
      <span>Total Marks</span>
      <b>{total} / {subjects.length * 100}</b>
    </div>
    <div>
      <span>Percentage</span>
      <b>{percentage.toFixed(2)}%</b>
    </div>
    <div>
      <span>Overall Grade</span>
      <b>{grade}</b>
    </div>
    <div>
      <span>Result</span>
      <b className="overall-pass">PASS</b>
    </div>
  </div>
  { /* Footer */
  <div className="card-footer">
    <p><b>Class Teacher</b></p>
    <p><b>Principal</b></p>
  </div>

</div>
</div>
);
}

```

```
export default Progress;
```

Progress.css

```
* {  
  box-sizing: border-box;  
}  
body {  
  margin: 0;  
  font-family: "Segoe UI", sans-serif;  
  background: linear-gradient(135deg, #667eea, #764ba2);  
}
```

```
/* MAIN PAGE */
```

```
.progress-page {  
  min-height: 100vh;  
  padding: 40px 20px;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
}
```

```
/* CARD */
```

```
.progress-card {  
  width: 900px;  
  padding: 35px;  
  background: #ffffff;  
  border-radius: 20px;  
  box-shadow: 0 15px 40px rgba(0, 0, 0, 0.2);  
}
```

```
/* COLLEGE HEADER */
```

```
.college-header {  
  text-align: center;  
  padding-bottom: 20px;  
  border-bottom: 1px solid #ddd;
```

```
}  
.college-header h1 {  
  margin: 5px 0;  
  font-size: 24px;  
  color: #4c3f91;  
}  
.college-header p {  
  margin: 8px 0;  
  color: #777;  
}  
.college-header h2 {  
  margin-top: 15px;  
  color: #222;  
}  
/* STUDENT DETAILS */  
.student-details {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
  gap: 10px 40px;  
  padding: 25px 10px;  
}  
.student-details p {  
  margin: 8px 0;  
  color: #444;  
}  
/* TABLE */  
table {  
  width: 100%;  
  border-collapse: separate;  
  border-spacing: 0;  
  margin-top: 10px;  
  overflow: hidden;  
  border-radius: 10px;
```

```

}

th,
td {
    padding: 14px;
    text-align: center;
    border-bottom: 1px solid #ddd;
}

th {
    background: #4c3f91;
    color: white;
    font-weight: 600;
}

td {
    background: #faf9ff;
}

td:nth-child(2) {
    text-align: left;
}

tr:hover td {
    background: #f0edff;
}

.pass {
    color: #219653;
    font-weight: bold;
}

/* SUMMARY */

.summary {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 15px;
    margin-top: 30px;
}

```

```
.summary div {
  padding: 20px 10px;
  text-align: center;
  background: #f4f1ff;
  border-radius: 12px;
  border: 1px solid #e2dcff;
}

.summary span {
  display: block;
  font-size: 13px;
  color: #777;
  margin-bottom: 10px;
}

.summary b {
  font-size: 20px;
  color: #4c3f91;
}

/* RESULT */
.overall-pass {
  color: #219653;
  font-weight: bold;
}

/* FOOTER */
.card-footer {
  display: flex;
  justify-content: space-between;
  margin-top: 60px;
  padding: 0 20px;
}

.card-footer p {
  min-width: 150px;
  text-align: center;
}
```

```
padding-top: 10px;  
border-top: 2px solid #aaa;  
color: #555;  
}
```

```
/* MOBILE */
```

```
@media (max-width: 700px) {  
  .progress-page {  
    padding: 15px;  
  }  
  .progress-card {  
    width: 100%;  
    padding: 20px;  
  }  
  .college-header h1 {  
    font-size: 18px;  
  }  
  .student-details {  
    grid-template-columns: 1fr;  
  }  
  table {  
    font-size: 12px;  
  }  
  th,  
  td {  
    padding: 9px;  
  }  
  .summary {  
    grid-template-columns: 1fr 1fr;  
  }  
  .card-footer {  
    padding: 0;  
    margin-top: 40px;  
  }  
}
```

```
    gap: 15px;
  }
  .card-footer p {
    min-width: 100px;
    font-size: 12px;
  }
}
```

App.jsx

```
import React from "react";
import Progress from "./Progress.jsx";
function App() {
  return (
    <Progress />
  );
}
export default App;
```

Output

PRINCE DR. K. VASUDEVAN COLLEGE OF ENGINEERING AND TECHNOLOGY

Department of Cyber Security

STUDENT PROGRESS CARD

Name:Flana

Register No: 411625104021

Department: B.E.

Semester: III

S.No	Subject	Marks	Grade	Result
1	Web Interface	85	A	PASS
2	Data Structures	78	B	PASS
3	Database Management	82	A	PASS
4	Java Programming	88	A	PASS
5	Computer Networks	75	B	PASS

Total Marks

408 / 500

Percentage

81.60%

Overall Grade

A

Result

PASS

Class Teacher

Principal

Result

The Student Progress Card application was successfully developed using React. The project demonstrates data-driven rendering with array methods, conditional grade logic, component-based architecture, and responsive CSS styling.